Generalized Limit Problem

Here is a rigorous mathematical generalization for the problem presented in the image `dailyintegral-limit-limit-1x1-2026-08-10.jpg`.

We can generalize the spacing from 1 to an arbitrary positive constant $a > 0$ and introduce a scaling factor in the denominator to preserve the 1^∞ indeterminate form.

The Generalized Problem

Let $a > 0$. We define the generalized function $g_a(x)$ as:

$$g_a(x) = \lim_{r \to 0} \left(\frac{(x+a)^{r+1} - x^{r+1}}{a} \right)^{\frac{1}{r}}$$

We want to find the generalized limit:

$$\lim_{x \to \infty} \frac{g_a(x)}{x}$$

Step 1: Evaluating the Inner Limit for $g_a(x)$

To evaluate $g_a(x)$, we note that substituting $r = 0$ yields 1^∞, which is an indeterminate form. We rewrite the expression using the exponential function:

$$g_a(x) = \exp\left(\lim_{r \to 0} \frac{\ln\left(\frac{(x+a)^{r+1} - x^{r+1}}{a} \right)}{r} \right)$$

Let L be the limit inside the exponential. As $r \to 0$, we have a $\frac{0}{0}$ indeterminate form, allowing us to apply L'Hôpital's Rule with respect to r:

$$L = \lim_{r \to 0} \frac{\frac{d}{dr}\left[\ln\left(\frac{(x+a)^{r+1} - x^{r+1}}{a} \right) \right]}{\frac{d}{dr}[r]}$$

Using the chain rule, the derivative of the numerator is:

$$\frac{d}{dr}\left[\ln\left(\frac{(x+a)^{r+1} - x^{r+1}}{a} \right) \right] = \frac{a}{(x+a)^{r+1} - x^{r+1}} \cdot \frac{d}{dr}\left[\frac{(x+a)^{r+1} - x^{r+1}}{a} \right]$$

Since $\frac{d}{dr}[c^{r+1}] = c^{r+1}\ln(c)$, we have:

$$\frac{d}{dr}\left[\frac{(x+a)^{r+1} - x^{r+1}}{a}\right] = \frac{(x+a)^{r+1}\ln(x+a) - x^{r+1}\ln(x)}{a}$$

Substituting this back and evaluating at $r = 0$:

$$L = \frac{(x+a)^1\ln(x+a) - x^1\ln(x)}{(x+a)^1 - x^1} = \frac{(x+a)\ln(x+a) - x\ln(x)}{a}$$

Thus, the generalized function simplifies to:

$$g_a(x) = \exp\left(\frac{(x+a)\ln(x+a) - x\ln(x)}{a}\right)$$

By properties of logarithms and exponentials, this can be written algebraically:

$$g_a(x) = \frac{(x+a)^{\frac{x+a}{a}}}{x^{\frac{x}{a}}}$$

Step 2: Evaluating the Limit to Infinity

Now we substitute $g_a(x)$ into the outer limit:

$$\lim_{x\to\infty} \frac{g_a(x)}{x} = \lim_{x\to\infty} \frac{(x+a)^{\frac{x+a}{a}}}{x \cdot x^{\frac{x}{a}}}$$

Combine the terms in the denominator:

$$\lim_{x\to\infty} \frac{(x+a)^{\frac{x}{a}+1}}{x^{\frac{x}{a}+1}} = \lim_{x\to\infty} \left(\frac{x+a}{x}\right)^{\frac{x}{a}+1}$$

Simplify the fraction:

$$\lim_{x\to\infty} \left(1 + \frac{a}{x}\right)^{\frac{x}{a}+1}$$

We can split this limit using exponent rules:

$$\lim_{x\to\infty} \left[\left(1 + \frac{a}{x}\right)^{\frac{x}{a}} \cdot \left(1 + \frac{a}{x}\right)^{1}\right]$$

To evaluate the first part, let $u = \frac{x}{a}$. As $x \to \infty$, $u \to \infty$:

$$\lim_{u\to\infty} \left(1 + \frac{1}{u}\right)^u = e$$

The second part simply evaluates to 1 as $x \to \infty$:

$$\lim_{x\to\infty} \left(1 + \frac{a}{x}\right) = 1 + 0 = 1$$

Conclusion

Multiplying the evaluated limits gives us the final result:

$$\lim_{x\to\infty} \frac{g_a(x)}{x} = e \cdot 1 = e$$

Notice that the limit evaluates identically to Euler's number e, regardless of the positive interval step size a. By setting $a = 1$, you yield the exact solution for the original expression in the image.